

Contrôle saturé pour certaines EDP : théorie et applications

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PDEs are everywhere

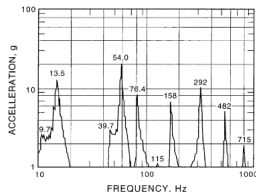
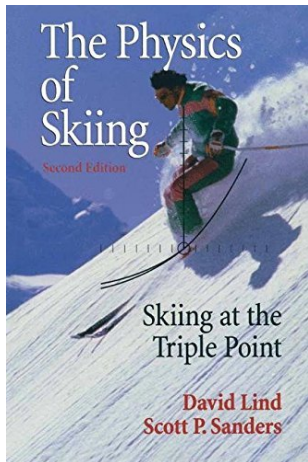
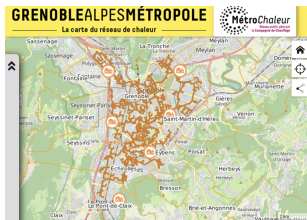


FIGURE 3.5. The frequency distribution of the normal vibration modes of a ski. The ski is clamped at the center to a shaker and driven. An output accelerometer located on the afterbody records the vibration response shown. [Reprinted with permission from R. L. Piziali and C. D. Mote, Jr., "The Snow Ski as a Dynamic System," J. Dynamic Syst. Meas. Control, Trans. ASME **94**, 134 (1972).]

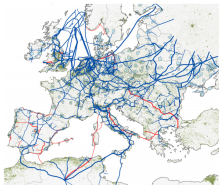
Page 63: Natural frequency with "good and bad vibrations"

[David A. Lind and Scott P. Sanders, *The Physics of Skiing*;2013]

PDEs are everywhere



Heating system [Grenoble]



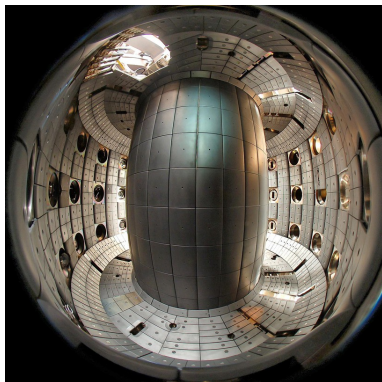
Gas network [Europe]



Water regulation [Sambre]



Traffic control [Nice]



Nuclear Fusion TCV
[Lausanne]

In most of these examples:

- (Non)linear PDEs and use of constrained control
- Often only boundary (or pointwise control)
- Linear control has to be overcome.

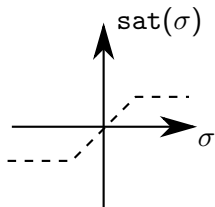
Experiments with PDE control at the end of this talk.

We are interested in computing control for linear and nonlinear PDEs:

$$\begin{aligned}\dot{z} &= \mathcal{F}(z, u) \\ z(0) &= z_0\end{aligned}\tag{1}$$

Given an equilibrium (say 0), compute a control u such that **asymptotic stability**:

- **stability**: For all $\varepsilon > 0$, there exists $\delta > 0$ such that if the initial condition $\|z_0\| \leq \delta$, it holds, for all $t \geq 0$, $\|z(t)\| \leq \varepsilon$
- **attractivity**: for all initial condition z_0 , it holds $z(t) \rightarrow 0$ as $t \rightarrow \infty$



$$u = \text{sat}(\sigma) = \begin{cases} \sigma & \text{if } |\sigma| < 1 \\ \text{sign}(\sigma) & \text{else} \end{cases}$$

Saturation can reduce the performance or even destabilize the system, even for finite-dimensional systems.

More precisely, even if

$$\dot{z} = Az + BKz$$

is asymp. stable, it may hold that

$$\dot{z} = Az + B\text{sat}(Kz)$$

is **not** globally asymptotically stable.

It may exist new equilibrium, new limit cycles...

See e.g. [Khalil; 1996] and [Tarbouriech, *et al*; 2011]

Goal of this talk:

What happens for some PDEs? as wave equation, or the nonlinear KdV equation, or parabolic PDE?

In presence of bounded and unbounded control operators?

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Two objectives

- Well-posedness
- Asymptotic stability

for different PDEs in presence of a distributed/boundary saturating control.

Review of different techniques and different results:

local vs global result

asymptotic/exponential/polynomial stability

control design results or analysis results

theory and real application

- 1 Well-posedness and stability of **wave equation** with a **saturated control**
- 2 Korteweg-de Vries equation with **saturated control**
- 3 Stabilization of **parabolic PDEs** in presence of saturations at the boundary control
Applications to **fusion control** and real experiments
- 4 Conclusion

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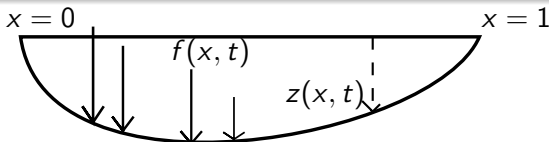
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Lyapunov method, LaSalle invariance principle
... or with a **saturated boundary control**
local sector bounded conditions
joint work with S. Tarbouriech and J.M. Gomes da Silva Jr
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1.a – Wave equation with a bounded control operator



1D wave equation with bounded control operator.

Dynamics of the vibration:

$$z_{tt}(x, t) = z_{xx}(x, t) + f(x, t), \quad \forall x \in (0, 1), t \geq 0, \quad (2)$$

Boundary conditions, $\forall t \geq 0$,

$$\begin{aligned} z(0, t) &= 0, \\ z(1, t) &= 0, \end{aligned} \quad (3)$$

and with the following initial condition, $\forall x \in (0, 1)$,

$$\begin{aligned} z(x, 0) &= z^0(x), \\ z_t(x, 0) &= z^1(x), \end{aligned} \quad (4)$$

where (z^0, z^1) stands for the initial condition

When closing the loop with a linear control

Let us define the **linear control** by

$$f(x, t) = -az_t(x, t), x \in (0, 1), \forall t \geq 0, \quad (5)$$

and consider

$$V_1 = \frac{1}{2} \int (z_x^2 + z_t^2) dx.$$

Formal computation. Along the solutions to (2), (3) and (5):

$$\begin{aligned} \dot{V}_1 &= \int_0^1 (z_x z_{xt} - az_t^2 + z_t z_{xx}) dx \\ &= - \int_0^1 az_t^2 dx + [z_t z_x]_{x=0}^{x=1} \\ &= - \int_0^1 az_t^2 dx \end{aligned}$$

Thus, if $a > 0$, V_1 is a (non strict) Lyapunov function.

Using standard technics (Lumer-Philipps thereom (for the well-posedness) and Huang-Prüss theorem (for the exp. stability)):

Proposition

$\forall a > 0, \forall (z^0, z^1) \text{ in } H_0^1(0, 1) \times L^2(0, 1),$
 $\exists ! \text{ solution } z: [0, \infty) \rightarrow H_0^1(0, 1) \times L^2(0, 1) \text{ to (2)-(5). Moreover,}$
 $\exists C, \mu > 0, \text{ such that, for any initial condition } H_0^1(0, 1) \times L^2(0, 1),$
it holds, $\forall t \geq 0,$

$$\|z\|_{H_0^1(0,1)} + \|z_t\|_{L^2(0,1)} \leq Ce^{-\mu t}(\|z^0\|_{H_0^1(0,1)} + \|z^1\|_{L^2(0,1)}).$$

In the previous proposition:

- stability
- attractivity of the equilibrium
- with an exponential speed

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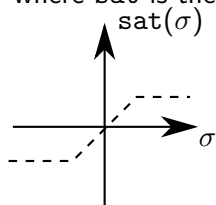
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- attractivity of the equilibrium
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When closing the loop with a saturating control

Let us consider now the **nonlinear control**

$$f(x, t) = -\text{sat}(az_t(x, t)), \quad x \in (0, 1), \quad \forall t \geq 0, \quad (6)$$

where sat is the localized saturated map:



$$\text{sat}(\sigma) = \begin{cases} \sigma & \text{if } |\sigma| < 1 \\ \text{sign}(\sigma) & \text{else} \end{cases}$$

Equation (2) in closed loop with the control (6) becomes

$$z_{tt} = z_{xx} - \text{sat}(az_t) \quad (7)$$

A formal computation gives, along the solutions to (7) and (3),

$$\dot{V}_1 = - \int_0^1 z_t \text{sat}(az_t) dx$$

which asks to handle the nonlinearity $z_t \text{sat}(az_t)$.

Remark: Choice of the saturation map

[Slemrod; 1989] and [Lasiecka and Seidman; 2003] deal with L^2 saturation (see also [Curtain and Zwart; 2016])

Given $\sigma : [0, 1] \rightarrow \mathbb{R}$, $\text{sat}_2(\sigma)$ is the function defined by

$$\text{sat}_2(\sigma)(x) = \begin{cases} \sigma(x) & \text{if } \|\sigma\|_{L^2(0,1)} < 1 \\ \frac{\sigma(x)}{\|\sigma\|_{L^2(0,1)}} & \text{else} \end{cases}$$

Here we consider **localized** saturation which is more physically relevant:

$$\text{sat}(\sigma(x)) = \begin{cases} \sigma(x) & \text{if } |\sigma(x)| < 1 \\ \text{sign}(\sigma(x)) & \text{else} \end{cases}$$

Well-posedness of this nonlinear PDE

Theorem [CP, Tarbouriech, Gomes da Silva Jr; 16]

$\forall a \geq 0$, for all (z^0, z^1) in $(H^2(0, 1) \cap H_0^1(0, 1)) \times H_0^1(0, 1)$, there exists a unique solution $z: [0, \infty) \rightarrow H^2(0, 1) \cap H_0^1(0, 1)$ to (7) with the boundary conditions (3) and the initial condition (4).

Consider

$$A_1 \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} v \\ u_{xx} - \text{sat}(av) \end{pmatrix}$$

with the domain $D(A_1) = (H^2(0,1) \cap H_0^1(0,1)) \times H_0^1(0,1)$.

Let us use a generalization of Lumer-Phillips theorem which is the so-called **Crandall-Liggett theorem**, as given in [Barbu; 1976]. See also [Brezis; 1973] and [Miyadera; 1992].

Again two conditions

- 1 A_1 is dissipative, that is

$$\Re \left(\left\langle A_1 \begin{pmatrix} u \\ v \end{pmatrix} - A_1 \begin{pmatrix} \tilde{u} \\ \tilde{v} \end{pmatrix}, \begin{pmatrix} u \\ v \end{pmatrix} - \begin{pmatrix} \tilde{u} \\ \tilde{v} \end{pmatrix} \right\rangle \right) \leq 0$$

- 2 For all $\lambda > 0$, $D(A_1) \subset \text{Ran}(I - \lambda A_1)$

First item: Easy step!

Instead of proving

$$\Re \left(\left\langle A_1 \begin{pmatrix} u \\ v \end{pmatrix} - A_1 \begin{pmatrix} \tilde{u} \\ \tilde{v} \end{pmatrix}, \begin{pmatrix} u \\ v \end{pmatrix} - \begin{pmatrix} \tilde{u} \\ \tilde{v} \end{pmatrix} \right\rangle \right) \leq 0, \text{ let us}$$

check, for all $\begin{pmatrix} u \\ v \end{pmatrix} \in H_0^1(0,1) \times L^2(0,1)$:

$$\Re \left(\left\langle A_1 \begin{pmatrix} u \\ v \end{pmatrix}, \begin{pmatrix} u \\ v \end{pmatrix} \right\rangle \right) \leq 0$$

To do that, using the definition of A_1 , and of the scalar product in $H_0^1(0,1) \times L^2(0,1)$, it is equal to:

$$\begin{aligned} & \int_0^1 v_x(x) \overline{u_x(x)} dx + \int_0^1 (u_{xx}(x) - \text{sat}(a v(x))) \overline{v(x)} dx, \\ &= \int_0^1 v_x(x) \overline{u_x(x)} dx + \int_0^1 u_{xx}(x) \overline{v(x)} dx - \int_0^1 \text{sat}(a v(x)) \overline{v(x)} dx \\ &= [u_x(x) \overline{v(x)}]_{x=0}^{x=1} - \int_0^1 \text{sat}(a v(x)) \overline{v(x)} dx \leq 0 \end{aligned}$$

due to the boundary and since $a \geq 0$.

Second item asks to deal with a nonlinear ODE.

Let $\begin{pmatrix} u \\ v \end{pmatrix} \in H_0^1(0,1) \times L^2(0,1)$ we have to find $\begin{pmatrix} \tilde{u} \\ \tilde{v} \end{pmatrix} \in D(A_1)$ such that

$$(I - \lambda A_1) \begin{pmatrix} \tilde{u} \\ \tilde{v} \end{pmatrix} = \begin{pmatrix} u \\ v \end{pmatrix}$$

that is

$$\begin{cases} \tilde{u} - \lambda \tilde{v} = u, \\ \tilde{v} - \lambda(\tilde{u}_{xx} - \text{sat}(a \tilde{v})) = v, \end{cases}$$

In particular, we have to find $\tilde{u}$ such that

$$\begin{aligned} \tilde{u}_{xx} - \frac{1}{\lambda^2} \tilde{u} - \text{sat}\left(\frac{a}{\lambda}(\tilde{u} - u)\right) &= -\frac{1}{\lambda} v - \frac{1}{\lambda^2} u \\ \tilde{u}(0) = \tilde{u}(1) &= 0 \end{aligned}$$

holds.

Nonhomogeneous nonlinear ODE with two boundary conditions

Lemma

If a is nonnegative and λ is positive, then there exists $\tilde{u}$ solution to

$$\begin{aligned}\tilde{u}_{xx} - \frac{1}{\lambda^2} \tilde{u} - \text{sat}\left(\frac{a}{\lambda}(\tilde{u} - u)\right) &= -\frac{1}{\lambda} v - \frac{1}{\lambda^2} u \\ \tilde{u}(0) &= \tilde{u}(1) = 0\end{aligned}\tag{8}$$

To prove this lemma, let us introduce the following map

$$\begin{aligned}\mathcal{T} : L^2(0,1) &\rightarrow L^2(0,1), \\ y &\mapsto z = \mathcal{T}(y),\end{aligned}$$

where $z = \mathcal{T}(y)$ is the unique solution to

$$\begin{aligned}z_{xx} - \frac{1}{\lambda^2} z &= -\frac{1}{\lambda} v - \frac{1}{\lambda^2} u + \text{sat}\left(\frac{a}{\lambda}(y - u)\right), \\ z(0) &= z(1) = 0.\end{aligned}$$

Prove that $\mathcal{T}$ is well defined and apply the Schauder fixed-point theorem (see e.g., [Coron; 2007]), to deduce that there exists y such that $\mathcal{T}(y) = y$

$\tilde{u} = y$ solves (8)

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Global asymptotic stability of this nonlinear PDE

Theorem

$\forall a > 0$, for all (z^0, z^1) in $(H^2(0, 1) \cap H_0^1(0, 1)) \times H_0^1(0, 1)$, the solution to (7) with the boundary conditions (3) and the initial condition (4) satisfies the following **stability property**, $\forall t \geq 0$,

$$\|z(\cdot, t)\|_{H_0^1(0,1)} + \|z_t(\cdot, t)\|_{L^2(0,1)} \leq \|z^0\|_{H_0^1(0,1)} + \|z^1\|_{L^2(0,1)} ,$$

together with the **attractivity property**

$$\|z(\cdot, t)\|_{H_0^1(0,1)} + \|z_t(\cdot, t)\|_{L^2(0,1)} \rightarrow 0, \text{ as } t \rightarrow \infty .$$

Due to theorem, the formal computation

$$\dot{V}_1 = - \int_0^1 z_t \text{sat}(az_t) dx$$

makes sense. This is only a weak Lyapunov function $\dot{V}_1 \leq 0$
(the state is (z, z_t) , and there is no $-z^2$).

To be able to apply **LaSalle's Invariance Principle**, we have to check that the trajectories are precompact (see e.g. [Dafermos, Slemrod; 1973], [d'Andréa-Novel *et al*; 1994]). It comes from:

Lemma

The canonical embedding from $D(A_1)$, equipped with the graph norm, into $H_0^1(0, 1) \times L^2(0, 1)$ is compact.

Sketch of the proof of

The canonical embedding from $D(A_1)$, equipped with the graph norm, into $H_0^1(0, 1) \times L^2(0, 1)$ is compact.

Consider a sequence $\begin{pmatrix} u_n \\ v_n \end{pmatrix}_{n \in \mathbb{N}}$ in $D(A_1)$, which is bounded with the graph norm, that is $\exists M > 0, \forall n \in \mathbb{N}$,

$$\begin{aligned} \left\| \begin{pmatrix} u_n \\ v_n \end{pmatrix} \right\|_{D(A_1)}^2 &:= \left\| \begin{pmatrix} u_n \\ v_n \end{pmatrix} \right\|^2 + \left\| A_1 \begin{pmatrix} u_n \\ v_n \end{pmatrix} \right\|^2, \\ &= \int_0^1 (|u_n'|^2 + |v_n|^2 + |v_n'|^2 \\ &\quad + |u_n'' - \text{asat}(v_n)|^2) dx < M \end{aligned}$$

From that, we deduce that $\int_0^1 (|v_n|^2 + |v_n'|^2) dx$ and $\int_0^1 (|u_n'|^2 + |u_n''|^2) dx$ are bounded.

Thus there exists a subsequence which converges in $H_0^1(0, 1) \times L^2(0, 1)$.



Using the dissipativity of A_1 , and previous lemma the trajectory $\begin{pmatrix} z(\cdot, t) \\ z_t(\cdot, t) \end{pmatrix}$ is precompact in $H_0^1(0, 1) \times L^2(0, 1)$.

Moreover the ω -limit set $\omega \left[\begin{pmatrix} z(\cdot, 0) \\ z_t(\cdot, 0) \end{pmatrix} \right] \subset D(A_1)$, is not empty and invariant with respect to the nonlinear semigroup $T(t)$ (see [Slemrod; 1989]).

We now use LaSalle's invariance principle to show that

$$\omega \left[\begin{pmatrix} z(\cdot, 0) \\ z_t(\cdot, 0) \end{pmatrix} \right] = \{0\}.$$

Therefore the convergence property holds. □

For linear PDE, we have exponential convergence

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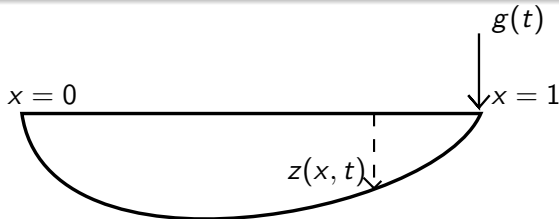
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For linear PDE, we have exponential convergence

1.b – Wave equation with a boundary control



1D wave equation with a boundary control.

Dynamics, $\forall x \in (0, 1), t \geq 0$,

$$z_{tt}(x, t) = z_{xx}(x, t), \quad (9)$$

Boundary conditions, $\forall t \geq 0$,

$$\begin{aligned} z(0, t) &= 0, \\ z_x(1, t) &= g(t), \end{aligned} \quad (10)$$

and with the same initial condition, $\forall x \in (0, 1)$,

$$\begin{aligned} z(x, 0) &= z^0(x), \\ z_t(x, 0) &= z^1(x). \end{aligned} \quad (11)$$

When closing the loop with a linear boundary control

Let us define the **linear control** by

$$g(t) = -bz_t(1, t), \quad x \in (0, 1), \quad \forall t \geq 0 \quad (12)$$

and consider (inspired by [Krstic; 2011])

$$V_2 = \frac{1}{2} \left(\int e^{\mu x} (z_t + z_x)^2 dx + \int e^{-\mu x} (z_t - z_x)^2 dx \right),$$

Formal computation. Along the solutions to (9), (10) and (12):

$$\dot{V}_2 = -\mu V_2 + \frac{1}{2} (e^{\mu}(1-b)^2 - e^{-\mu}(1+b)^2) z_t^2(1, t)$$

Assuming $b > 0$ and letting $\mu > 0$ such that $e^{\mu}(1-b)^2 \leq e^{-\mu}(1+b)^2$, it holds $\dot{V}_2 \leq -\mu V_2$ and thus V_2 is a strict Lyapunov function and thus (9)-(12) is exponentially stable.

When closing the loop with a linear boundary control

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$$g(t) = -bz_t(1, t), \quad x \in (0, 1), \quad \forall t \geq 0 \quad (12)$$

and consider (inspired by [Krstic; 2011])

$$V_2 = \frac{1}{2} \left(\int e^{\mu x} (z_t + z_x)^2 dx + \int e^{-\mu x} (z_t - z_x)^2 dx \right),$$

Formal computation. Along the solutions to (9), (10) and (12):

$$\dot{V}_2 = -\mu V_2 + \frac{1}{2} (e^{\mu}(1-b)^2 - e^{-\mu}(1+b)^2) z_t^2(1, t)$$

Assuming $b > 0$ and letting $\mu > 0$ such that

$e^{\mu}(1-b)^2 \leq e^{-\mu}(1+b)^2$, it holds $\dot{V}_2 \leq -\mu V_2$ and thus V_2 is a strict Lyapunov function and thus (9)-(12) is exponentially stable.

When closing the loop with a saturating control

Let us consider now the **nonlinear control**

$g(t) = -\text{sat}(bz_t(1, t)), \forall t \geq 0$. The boundary conditions become:

$$z(0, t) = 0, \quad z_x(1, t) = -\text{sat}(bz_t(1, t)). \quad (13)$$

Theorem

$\forall b > 0$, for all (z^0, z^1) in $\{(u, v), (u, v) \in H^2(0, 1) \times H_{(0)}^1(0, 1), u_x(1) + \text{sat}(bv(1)) = 0, u(0) = 0\}$, the solution to (9) with the boundary conditions (13) and the initial condition (4) satisfies the following **stability property**, $\forall t \geq 0$,

$$\|z(\cdot, t)\|_{H_{(0)}^1(0,1)} + \|z_t(\cdot, t)\|_{L^2(0,1)} \leq \|z^0\|_{H_{(0)}^1(0,1)} + \|z^1\|_{L^2(0,1)},$$

together with the **attractivity property**

$$\|z(\cdot, t)\|_{H_{(0)}^1(0,1)} + \|z_t(\cdot, t)\|_{L^2(0,1)} \rightarrow 0, \quad \text{as } t \rightarrow \infty.$$

To prove the well-posedness of the Cauchy problem we prove that A_2 defined by

$$A_2 \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} v \\ u'' \end{pmatrix}$$

with the domain $D(A_2) = \{(u, v), (u, v) \in H^2(0, 1) \times H_{(0)}^1(0, 1), u'(1) + \text{sat}(bv(1)) = 0, u(0) = 0\}$ is a semigroup of contraction.

The global stability property comes directly from the dissipativity of A_2 .

See also [Hastir, Califano and Zwart; 2019] for the well-posedness.

The global attractivity property comes from the following lemma:

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The **global stability property** comes directly from the dissipativity of A_2 .

See also [Hastir, Califano and Zwart; 2019] for the well-posedness.

The **global attractivity property** comes from the following lemma:

Lemma (semi-global exponential stability)

For all $r > 0$, there exists $\mu > 0$ such that, for all initial conditions satisfying

$$\|z^{0''}\|_{L^2(0,1)}^2 + \|z^1\|_{H_{(0)}^1(0,1)}^2 \leq r^2, \quad (14)$$

it holds

$$\dot{V}_2 \leq -\mu V_2$$

along the solutions to (9) with the boundary conditions (13).

Sketch of the proof of this lemma

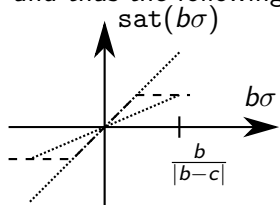
First note that by dissipativity of A_2 , it holds that

$$t \mapsto \left\| A_2 \begin{pmatrix} z(\cdot, t) \\ z_t(\cdot, t) \end{pmatrix} \right\|$$

is a non-increasing function. Now for all initial conditions satisfying (14), there exists $c \neq b$ such that, for all $t \geq 0$,

$$(b - c)|z_t(1, t)| \leq 1$$

and thus the following **local sector condition** holds:



Letting $\sigma = z_t(1, t)$, it holds
 $(\text{sat}(b\sigma) - b\sigma)(\text{sat}(b\sigma) - (b - c)\sigma) \leq 0$

We come back to the Lyapunov function candidate V_2 . Given $b > 0$, using the previous inequality, we compute

$$\begin{aligned}\dot{V}_2 &= -\mu V_2 + e^\mu (\sigma - \text{sat}(b\sigma))^2 - e^{-\mu} (\sigma + \text{sat}(b\sigma))^2 \\ &\leq -\mu V_2 + \begin{pmatrix} \sigma \\ \text{sat}(b\sigma) \end{pmatrix}^\top \begin{pmatrix} e^\mu - e^{-\mu} - b^2(b-c) & -e^\mu - e^{-\mu} + b + b(b-c) \\ -e^\mu - e^{-\mu} + b + b(b-c) & -1 + e^\mu - e^{-\mu} \end{pmatrix} \\ &\quad \times \begin{pmatrix} \sigma \\ \text{sat}(b\sigma) \end{pmatrix} \\ &\leq -\mu V_2\end{aligned}$$

with a suitable choice of constant values μ and c .

The semi-global exponential stability follows. □

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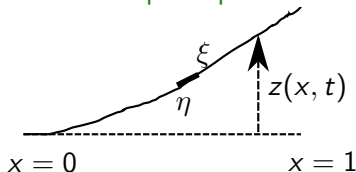
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with a suitable choice of constant values μ and c .

The semi-global exponential stability follows. □

Beam equation with a piezo patch

1D beam equation with a piezo patch:



Dynamics of the vibration, $\forall x \in (0, 1), t \geq 0$,

$$z_{tt}(x, t) + z_{xxxx}(x, t) = u(t) d_x[\delta_\eta(x) - \delta_\xi(x)], \quad (15)$$

where δ_η and δ_ξ are two Dirac measures

and $u(t)$ is the input (proportional to a controlled voltage)

It is a pointwise control through an unbounded control operator.

Assume that the beam is clamped at one end and free at the other end i.e. we consider the following **boundary conditions**

$$z(0, t) = z_x(0, t) = z_{xx}(1, t) = z_{xxx}(1, t) = 0 \quad t \geq 0 \quad (16)$$

Initial condition

$$z(x, 0) = z^0(x), \quad z_t(x, 0) = z^1(x) \quad 0 < x < 1 \quad (17)$$

where z^0 and z^1 stand resp. for the initial deflection and initial deflection speed.

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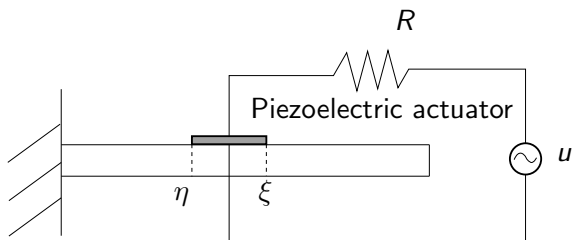
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where z^0 and z^1 stand resp. for the initial deflection and initial deflection speed.

Same approach works!

Joint work with [Andrea Mattioni](#) and [Sophie Tarbouriech](#)

also with dynamic control modelling



See also recent work of Lassi Paunonen.

1.c – Polynomial stability

Polynomial stability is even possible.

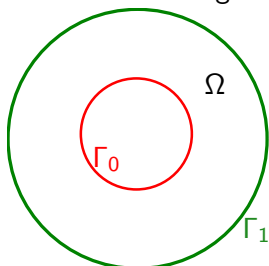
Let us consider the 2-D case

$$z_{tt}(x, t) = \Delta z, \quad \forall (x, t) \in \Omega \times [0, \infty) \quad (18a)$$

$$z(x, t) = 0, \quad \forall (x, t) \in \Gamma_0 \times [0, \infty) \quad (18b)$$

$$z(x, t) = \text{sat}(u), \quad \forall (x, t) \in \Gamma_1 \times [0, \infty) \quad (18c)$$

with the following "donut" geometry



Note the Dirichlet action,
and no more Neumann action.

See the related work of F. Alabau-Boussouira, in particular
[F. Alabau-Boussouira; 2012]

Deriving the classical norm in $H^1(\Omega) \times L^2(\Omega)$ gives:

$$\frac{1}{2} d_t \int_{\Omega} (\nabla z^2 + z_t^2) dx = \int_{\Gamma_0 \cup \Gamma_1} \partial_{\nu} z z_t ds$$

that does not help with our boundary control

However considering the weaker norm in $L^2(\Omega) \times H^{-1}(\Omega)$ defined by

$$\mathcal{E} = \frac{1}{2} \int_{\Omega} (z^2 + (A^{-1/2} z_t)^2) dx \text{ where } A = (-\Delta \text{ with zero Dirichlet b.c.})$$

gives

$$\dot{\mathcal{E}} = - \int_{\Gamma_0 \cup \Gamma_1} z \partial_{\nu} [A^{-1} z_t] ds = - \int_{\Gamma_1} \text{sat}(u) \partial_{\nu} [A^{-1} z_t] ds$$

This motivates the following definition for the control

$$u = \partial_{\nu} [A^{-1} z_t]$$

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Closed-loop system

$$z_{tt}(x, t) = \Delta z, \quad \forall (x, t) \in \Omega \times [0, \infty) \quad (19a)$$

$$z(x, t) = 0, \quad \forall (x, t) \in \Gamma_0 \times [0, \infty) \quad (19b)$$

$$z(x, t) = \text{sat}(\partial_\nu[A^{-1}z_t]), \quad \forall (x, t) \in \Gamma_1 \times [0, \infty) \quad (19c)$$

Theorem [Vanspranghe, Ferrante, CP; 2023]

The closed-loop PDE generates a strongly continuous semigroup of contraction in the $L^2(\Omega) \times H^{-1}(\Omega)$.

The closed-loop system is globally asymptotically stable

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The closed-loop PDE generates a strongly continuous semigroup of contraction in the $L^2(\Omega) \times H^{-1}(\Omega)$.

The closed-loop system is globally asymptotically stable

Inspired by [Komornik; 1994], using the multiplier method, we may prove the following **polynomial decay rate**:

Theorem [Vanspranghe, Ferrante, CP; 2023]

For any initial condition (z_0, z_1) in $D(\mathcal{A})$, there exists $C = C(\|\mathcal{A}(z_0, z_1)\|_{H^1(\Omega) \times L^2(\Omega)})$ such that, for all $t \geq 0$

$$\mathcal{E}(z(t), z_t(t)) \leq \frac{C}{\sqrt{t}}$$

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Lyapunov method, LaSalle invariance principle
... or with a saturated boundary control
local sector bounded conditions
- 2 Korteweg-de Vries equation with saturated in-domain control
contradiction argument
(boundary control is also possible)
joint work with Eduardo Cerpa
- 3 Stabilization of parabolic PDEs in presence of saturations at the boundary control
Applications to fusion control and real experiments
- 4 Conclusion and open questions

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3 – KdV equation

Let us consider the following nonlinear PDE

$$\begin{cases} z_t + z_x + z_{xxx} + zz_x + f = 0, & x \in [0, L], \ t \geq 0, \\ z(0, t) = z(L, t) = z_x(L, t) = 0, & t \geq 0, \\ z(x, 0) = z^0(x), & x \in [0, L], \end{cases} \quad (\text{KdV})$$

where z stands for the state and f for the **control**.

As noted in [Rosier; 1997], if $f = 0$ and

$$L \in \left\{ 2\pi \sqrt{\frac{k^2 + kl + l^2}{3}} / k, l \in \mathbb{N}^* \right\},$$

then, there exist solutions of the linearized version of (KdV) for which the $L^2(0, L)$ -energy does not decay to zero. Consider, for instance, $L = 2\pi$ and $z^0 = 1 - \cos(x)$ for all $x \in [0, L]$.

See [Coron, Crépeau; 2004], [Cerpa; 2007] and [Cerpa, Crépeau; 2009] for controllability properties with critical lengths.

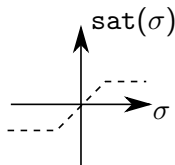
See also [Tang, Krstic; 2013] on linearized KdV equation with anti-diffusion.

Saturating control for KdV

Given $a > 0$, let us consider the KdV equation controlled by a saturated distributed control as follows

$$\begin{cases} z_t + z_x + z_{xxx} + zz_x + \text{sat}(az) = 0, \\ z(0, t) = z(L, t) = 0, \\ z_x(L, t) = 0, \\ z(x, 0) = z^0(x). \end{cases} \quad (\text{KdVsat})$$

where sat is the same saturation as before:



$$\text{sat}(\sigma) = \begin{cases} \sigma & \text{if } |\sigma| < 1 \\ \text{sign}(\sigma) & \text{else} \end{cases}$$

Theorem: well posedness [Marx, Cerpa, CP, Andrieu; 2016]

For any initial condition $z^0 \in L^2(0, L)$, there exists a unique solution $z \in C([0, T]; L^2(0, L)) \cap L^2(0, T; H^1(0, L))$ to (KdVsat).

Sketch of proof

First we prove the existence of solution on $[0, T']$ for T' small, as done in [Rosier, Zhang; 2006] and [Chapouly; 2009].

Then we remove the smallness assumption on T' by proving an a priori estimate.

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Then we remove the smallness assumption on T' by proving an a priori estimate.

Theorem: global asymptotic stability

There exist μ and a class $\mathcal{K}$ function $\alpha : [0, \infty) \rightarrow [0, \infty)$ such that for any $z^0 \in L^2(0, 1)$, any solution z to (KdVsat) satisfies, for all $t \geq 0$,

$$\|z(t)\|_{L^2(0,1)} \leq \alpha(\|z^0\|_{L^2(0,1)})e^{-\mu t}.$$

(See [Parada; 2022] for saturating boundary control)

To prove the result, we follow the approach of [Rosier, Zhang; 2006] and [Cerpa; 2014].

To be more specific:

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To prove the result, we follow the approach of [Rosier, Zhang; 2006] and [Cerpa; 2014].

To be more specific:

Lemma: semi-global exponential stability

Given $r > 0$, there exist positive values C and μ such that for all initial condition z^0 satisfying $\|z^0\|_{L^2(0,L)} \leq r$, it holds, for all $t \geq 0$,

$$\|z(t)\|_{L^2(0,L)} \leq C \|z^0\|_{L^2(0,L)} e^{-\mu t}.$$

To do that, the key result is:

$\forall T > 0, r > 0, \exists C > 0$ such that for any solution z to (KdVsat) starting from $z^0 \in L^2(0,L)$ with $\|z^0\|_{L^2(0,L)} \leq r$, it holds

$$\|z^0\|_{L^2(0,L)}^2 \leq C \left(\int_0^T |z_x(0,t)|^2 dt + 2 \int_0^T \int_0^L \text{sat}(az)z dt dx \right). \quad (20)$$

Assume (20) for the time being. Then with

$$\begin{aligned}\|z(\cdot, T)\|_{L^2(0,L)}^2 &= \|z^0\|_{L^2(0,L)}^2 - \int_0^T |z_x(0,t)|^2 dt \\ &\quad - 2 \int_0^T \int_0^L \text{sat}(az)z \, dx \, dt\end{aligned}$$

we get

$$\|z(\cdot, kT)\|_{L^2(0,L)}^2 \leq \gamma^k \|z^0\|_{L^2(0,L)}^2 \quad \forall k \geq 0$$

where $\gamma \in (0, 1)$. From the dissipativity property, we have $\|z(\cdot, t)\|_{L^2(0,L)} \leq \|z(\cdot, kT)\|_{L^2(0,L)}$ for $kT \leq t \leq (k+1)T$. Thus we obtain, for all $t \geq 0$,

$$\|z(\cdot, t)\|_{L^2(0,L)}^2 \leq \frac{1}{\gamma} \|z^0\|_{L^2(0,L)}^2 e^{\frac{\log \gamma}{T} t}$$

We conclude the proof of the semi-global exponential stability. $\square$

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$$\begin{aligned}\|z(\cdot, T)\|_{L^2(0,L)}^2 &= \|z^0\|_{L^2(0,L)}^2 - \int_0^T |z_x(0,t)|^2 dt \\ &\quad - 2 \int_0^T \int_0^L \text{sat}(az)z \, dx \, dt\end{aligned}$$

we get

$$\|z(\cdot, kT)\|_{L^2(0,L)}^2 \leq \gamma^k \|z^0\|_{L^2(0,L)}^2 \quad \forall k \geq 0$$

where $\gamma \in (0, 1)$. From the dissipativity property, we have

$\|z(\cdot, t)\|_{L^2(0,L)} \leq \|z(\cdot, kT)\|_{L^2(0,L)}$ for $kT \leq t \leq (k+1)T$. Thus we obtain, for all $t \geq 0$,

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Proof of the blue equation (20). By contradiction !

Assume that there exists a sequence of solution z^n to (KdVsat) with

$$\|z^n(\cdot, 0)\|_{L^2(0,L)} \leq r \quad (21)$$

and such that

$$\lim_{n \rightarrow +\infty} \frac{\|z^n\|_{L^2(0,T;L^2(0,L))}^2}{\int_0^T |z_x^n(0,t)|^2 dt + 2 \int_0^T \int_0^L \text{sat}(az^n(x,t)) z^n(x,t) dt dx} = +\infty. \quad (22)$$

By dissipativity property, $\exists \beta > 0$,

$$\sup_{t \in [0,T]} \|z^n(\cdot, t)\|_{L^2(0,L)} \leq r, \quad \sup_{x \in [0,L]} \int_0^T |z^n(x,t)|^2 dt \leq \beta. \quad (23)$$

Now let us define $\Omega_i := \left\{ t \in [0, T], \sup_{x \in [0,L]} |z(x,t)| > i \right\}$, it holds

$$\beta \geq \int_0^T \sup_{x \in [0,L]} |z^n(x,t)|^2 dt \geq \int_{\Omega_i} \sup_{x \in [0,L]} |z^n(x,t)|^2 dt \geq i^2 \nu(\Omega_i),$$

Therefore, we obtain for the complementary set

$$\nu(\Omega_i^c) \geq \max\left(T - \frac{\beta}{i^2}, 0\right).$$

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Therefore, we obtain for the complementary set

$$\nu(\Omega_i^c) \geq \max\left(T - \frac{\beta}{i^2}, 0\right).$$

Moreover, using again the **local sector condition**, we have, for a suitable positive function k and $\forall i \in \mathbb{N}$,

$$\begin{aligned} \int_0^T \int_0^L \text{sat}(az^n) z^n \, dt \, dx &= \int_{\Omega_i^c} \int_0^L \text{sat}(az^n) z^n \, dt \, dx + \int_{\Omega_i} \int_0^L \dots \\ &\geq \int_{\Omega_i^c} \int_0^L ak(i)(z^n)^2 \, dt \, dx + 0. \end{aligned} \quad (24)$$

Let $\lambda^n := \|z^n\|_{L^2(0,T;L^2(0,L))}$ and $v^n(x,t) = \frac{z^n(x,t)}{\lambda^n}$. Due to (21), up to extracting a subsequence, we may assume that $\lambda^n \rightarrow \lambda \geq 0$. Due to (22) and (24), we have, for all $i \in \mathbb{N}$

$$\int_0^T |v_x^n(0,t)|^2 dt + 2 \int_{\Omega_i^c} \int_0^L ak(i)(v^n)^2 dt dx \rightarrow 0 \quad (25)$$

Using Aubin-Lions lemma in [Simon, 1987]: $\{v^n\}_{n \in \mathbb{N}}$ converges strongly in $L^2(0,T;L^2(0,L))$. Thus, with (25), we have, for all $i \in \mathbb{N}$

$$v_x(0,t) = 0, \forall t \in (0,T) \text{ and } v(x,t) = 0, \forall x \in [0,L], \forall t \in \Omega_i^c.$$

We know that $\nu \left(\bigcup_{i \in \mathbb{N}} \Omega_i^c \right) = T$. We get a contradiction with $\|v\|_{L^2(0,T;L^2(0,L))} = 1$. □

Outline

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- 3 Stabilization of **parabolic PDEs...**
Sturm Liouville theory
Kalman conditions for observability and control
... in presence of saturations
Use of sector conditions
joint work with Hugo Lhachemi
Applications to fusion control
Real experiments
joint work with Bojan Mavkov and Emmanuel Witrant
- 4 Conclusion and open questions

3.a – Parabolic PDE

Consider the **reaction-diffusion** system:

$$z_t(x, t) = (p(x)z_x(x, t))_x + (q_c - q(x))z(x, t) \quad (26a)$$

$$z_x(0, t) = 0, \quad z(1, t) = \text{sat}(u(t)) \quad (26b)$$

$$z(x, 0) = z_0(x) \quad (26c)$$

$$y(t) = z(0, t) \quad (26d)$$

where

- $q_c \in \mathbb{R}$, $q \geq 0 \in \mathcal{C}^0$ and $p > 0 \in \mathcal{C}^2$ are two functions;
- $u \in \mathbb{R}$ is the control;
- $y \in \mathbb{R}$ is the output;
- $z_0 \in L^2(0, 1)$ is the initial condition;
- $z(\cdot, t) \in L^2(0, 1)$ is the state at time t .

Control objective

Derive stabilizing output feedback law?

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where we are interested in the heat equation

- $u \in \mathbb{R}$ is the control; controlled flux temperature
- $y \in \mathbb{R}$ is the output; mesure of a boundary temperature
- $z_0 \in L^2(0, 1)$ is the initial condition; distribution temperature at $t = 0$
- $z(\cdot, t) \in L^2(0, 1)$ is the state at time t .

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Use of finite-dimensional controllers for diffusion systems

[Balas; 1988], [Curtain; 1982] and [Sakawa, *Feedback stabilization of linear diffusion systems*; 1983], with the construction of a finite dimensional feedback dynamic controller for an arbitrarily large damping constant.

For either a boundary measure or a boundary actuation:
[Katz, Fridman; 2020]

Design method

- **State-feedback** is performed on the N_0 first modes
- **Observer** to estimate the N first modes, $N \geq N_0$
- Study of the **closed-loop system**

In the linear case

Inspired by [Curtain, Zwart; 20], we introduce the change of variable:

$$w(x, t) = z(x, t) - x^2 u(t). \quad (27)$$

Then we have

$$\begin{aligned} w_t(x, t) &= (p(x)w_x(x, t))_x + (q_c - q(x))w(x, t) \\ &\quad + a(x)u(t) + b(x)\dot{u}(t) \end{aligned} \quad (28a)$$

$$w_x(0, t) = 0, \quad w(1, t) = 0 \quad (28b)$$

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with (explicit) $a, b \in L^2(0, 1)$

Spectral reduction

In the linear case

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$$w_x(0, t) = 0, \quad w(1, t) = 0 \quad (28b)$$

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with (explicit) $a, b \in L^2(0, 1)$ defined by, for all x in $(0, 1)$, $a(x) = 2p(x) + 2xp'(x) + (q_c - q(x))x^2$, $b(x) = -x^2$, and $w_0(x) = z_0(x) - x^2 u(0)$.

Sturm Liouville theory

Let $\mathcal{A}$ be defined by $\mathcal{A}(f)(x) = -d_x(p(x)d_x f) + qf$ with domain $D(\mathcal{A}) = \{f \in H^2(0,1) : f'(0) = f(1) = 0\}$.

The eigenvalues λ_n , $n \geq 1$, of $\mathcal{A}$ are simple and increasing.

Moreover, for n sufficiently large, $n^2 p_* \leq \lambda_n \leq n^2 p^*$.

The unit eigenvectors $\Phi_n \in L^2(0,1)$ form a Hilbert basis and are such that $\Phi_n(0) = O(1)$ and $\Phi'_n(0) = O(\sqrt{\lambda_n})$ as $n \rightarrow +\infty$.

In particular there are **a finite number of unstable eigenvalues**.

$$N_0 := \# \text{ of modes such that } -\lambda_n + q_c \geq -\delta$$

Results in the linear case [Katz, Fridman; 20] & [Lhachemi, CP; 22]

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Sector nonlinearity constraint

In the nonlinear case, consider

$$z_t(x, t) = (p(x)z_x(x, t))_x + (q_c - q(x))z(x, t) \quad (29a)$$

$$z_x(0, t) = 0, \quad z(1, t) = \varphi(u(t)) =: u_\varphi(t) \quad (29b)$$

$$z(x, 0) = z_0(x) \quad (29c)$$

$$y(t) = z(0, t) \quad (29d)$$

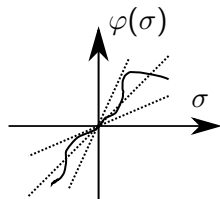
where φ is a smooth saturation map:

$\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is $\mathcal{C}^1$ satisfying $\exists k_\varphi > 0$ and

$\Delta k_\varphi \in (0, k_\varphi)$ s.t.

$$(k_\varphi - \Delta k_\varphi)|\sigma| \leq \text{sign}(\sigma)\varphi(\sigma) \leq (k_\varphi + \Delta k_\varphi)|\sigma|$$

for all $\sigma \in \mathbb{R}$.



With the same change of variable as at the beginning:

$$v_\varphi(t) = \dot{u}_\varphi(t) = \varphi'(u(t))\dot{u}(t) \quad (30a)$$

$$w_t(x, t) = (p(x)w_x(x, t))_x - (q - q_c)w(x, t) + a(x)u_\varphi(t) + b(x)v_\varphi(t) \quad (30b)$$

$$w_x(0, t) = 0, \quad w_x(1, t) = 0 \quad (30c)$$

$$w(0, x) = w_0(x) \quad (30d)$$

$$y(t) = w(0, t) \quad (30e)$$

The projection yields

$$\dot{u}_\varphi = v_\varphi = \varphi'(u)\dot{u} \quad (31a)$$

$$\dot{w}_n(t) = (-\lambda_n + q_c)w_n(t) + a_n u_\varphi(t) + b_n v_\varphi(t), \quad n \geq 1 \quad (31b)$$

$$y(t) = \sum_{n \geq 1} w_n(t) \phi_n(0) \quad (31c)$$

φ is not injective, thus the **red equation** can not be used for the design of u .

Let $\delta > 0$ and $N_0 \geq 1$ be such that $-\lambda_n + q_c < -\delta < 0$ for all $n \geq N_0 + 1$. We consider the following observer-based control strategy:

$$\hat{w}_n(t) = \hat{z}_n(t) + b_n u_\varphi(t) \quad (32a)$$

$$\dot{\hat{z}}_n(t) = (-\lambda_n + q_c)\hat{z}_n(t) + \beta_n u_\varphi(t) \quad (32b)$$

$$- l_n \left\{ \sum_{k=1}^N \hat{w}_k(t) \phi_k(0) - y(t) \right\}, \quad 1 \leq n \leq N_0$$

$$\dot{\hat{z}}_n(t) = (-\lambda_n + q_c)\hat{z}_n(t) + \beta_n u_\varphi(t), \quad N_0 + 1 \leq n \leq N \quad (32c)$$

$$u(t) = \sum_{k=1}^{N_0} k_k \hat{z}_k(t) \quad (32d)$$

Here $l_n \in \mathbb{R}$ and $k_k \in \mathbb{R}$ are the observer and feedback gains, respectively.

Existence of solutions to (31) and (32) follows from classical results. See Chapter 11 of [Curtain and Zwart; 2020].

Some remarks follow

- Since the system is open loop unstable, it is not possible to design a saturating stabilizing controller ✗
- Estimation of the basin of attraction with a Lyapunov function candidate ✓
- The Kalman controllability and observability conditions hold ✓

Using an appropriate scaling of the state variable and Lyapunov function we may get the following:

Design of a stabilizing output feedback law

Theorem: [Lhachemi, CP; 21], see also [Mironchenko, CP, and Wirth; 21]

Let $K \in \mathbb{R}^{1 \times N_0}$ and $L \in \mathbb{R}^{N_0}$ be such that $A_1 + B_1 K$ and $A_0 + LC_0$ are Hurwitz with real part $< -\delta < 0$.

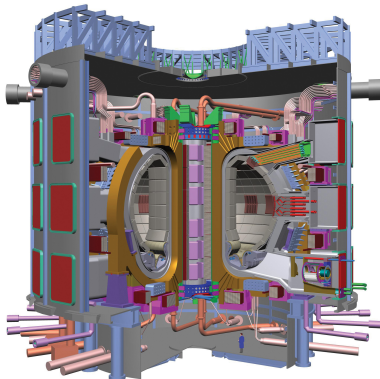
Then there exist $N \geq N_0 + 1$, and $P \succeq 0$, γ, μ and $M, \kappa > 0$ such that for all initial condition in

$$\mathcal{E}_1 = \left\{ Z \in L^2(0, 1) \times \mathbb{R} : \begin{bmatrix} u \\ \pi_{NZ} \end{bmatrix}^\top P \begin{bmatrix} u \\ \pi_{NZ} \end{bmatrix} + \gamma \|\mathcal{R}_{NZ}\|_{L^2}^2 \leq \frac{1}{\mu} \right\}.$$

for the parabolic PDE and any initial condition of the observer, $\hat{z}_n(0)$ for all $1 \leq n \leq N$, the solution satisfies, for all $t \geq 0$.

$$\|Z(t, \cdot)\|_{L^2 \times \mathbb{R}}^2 + u(t)^2 + \sum_{n=1}^N \hat{z}_n(t)^2 \leq M e^{-2\kappa t} (\|z_0\|_{L^2}^2 + u_0^2 + \sum_{n=1}^N \hat{z}_n(0)^2)$$

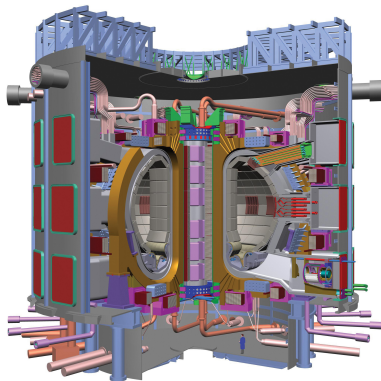
3.b – Application to fusion control



Fusion: Decarbonized energy
and little high-level nuclear
waste

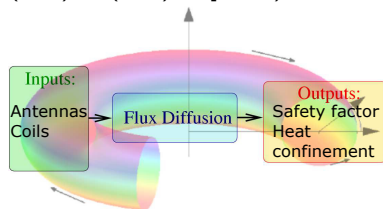
Dynamics in a torus:
 $(x, t) \in (0, 1) \times [0, \infty)$

3.b – Application to fusion control



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The electron temperature T in Tokamak plasma is "modelled" as,
 $(x, t) \in (0, 1) \times [0, \infty)$,

$$\begin{aligned}\partial_t T(x, t) &= \partial_x(p(T, x)\partial_x T)(x, t) + P_{sinks}(T) + P_{sources} , \\ \partial_x T(0, t) &= 0, \quad T(1, t) = T_{edge}(t) ,\end{aligned}$$

where P_{sinks} is a nonlinear function, p is a state-dependent diffusion, and $P_{sources}$ is the control
See [F. Bribiesca Argomedo, PhD], [B. Mavkov, PhD] for model.

Control problem

Stabilize around a given equilibrium with control $P_{sources}$, where nonlinearities (as saturation) are considered

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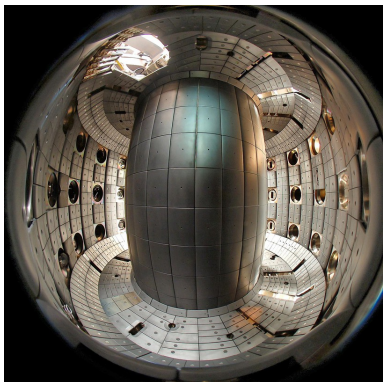
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Experimental results

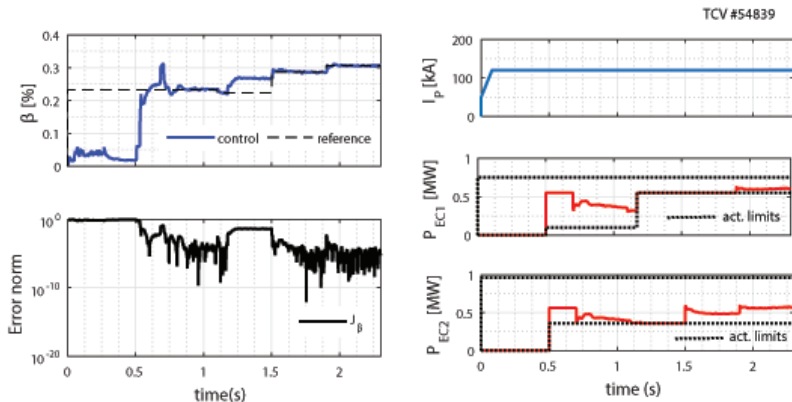


Experimental results have been done on TCV (Lausanne) joint work with [B. Mavkov](#), E. Witrant, E. Maljaars, F. Felici, O. Sauter and the TCV-team

See [B. Mavkov, *et al*, *Nuclear Fusion*; 2018].

Example of shocks that have been performed focusing on the β variable only

Particular shock with saturated control

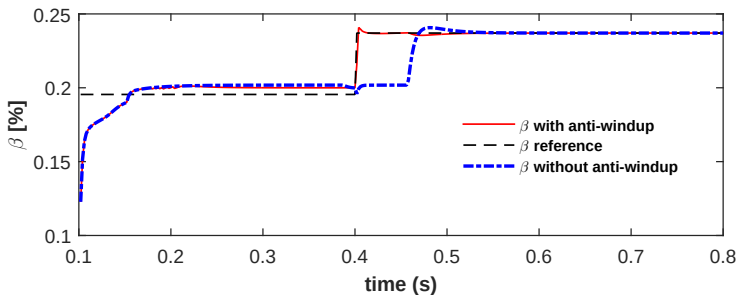


reference (—),
closed-loop (—)
and error (—)

saturated controls (— and —)
and actuator limits (···)

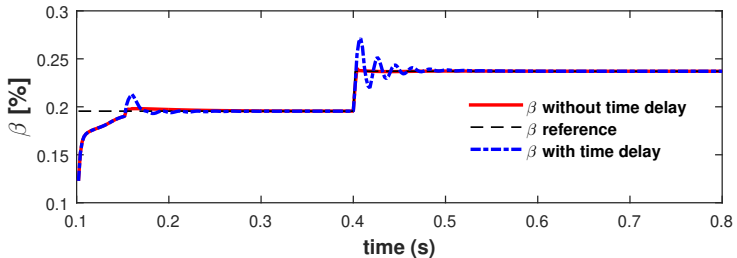
anti-windup loop

To counter-act the saturation maps: use of anti-windup loops.



with anti-windup loop (—)
without anti-windup loop (— · —)
reference (— —),

Delay in the loops



with delay in the loop (— · —)
without delay in the loop (—)
reference (— —),

- 1 Well-posedness and stability of wave equation with a saturated in-domain control
... or with a saturated boundary control
- 2 Korteweg-de Vries equation with saturated in-domain control
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- 3 Stabilization of parabolic PDEs...
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4 – Conclusion and open questions

Conclusion

- Well-posedness and global asymptotic stability for the linear PDEs and nonlinear PDEs.

In particular the following wave

$$\begin{cases} z_{tt}(x, t) - z_{xx}(x, t) = \text{sat}(u) \\ \text{boundary conditions} \\ \text{dynamical feedback controller} \end{cases}$$

- non-strict Lyapunov functions
- local sector conditions
- both the input and the output may saturate
- constructive methods and non-constructive ones

4 – Conclusion and open questions

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- both the input and the output may saturate
- constructive methods and non-constructive ones

Also studied

- Other equations as **beam equation** and pointwise control
- **Nonlinear control** and **additional dynamics** could be considered as well
- coupling of PDEs and ODEs
- other methods for the 2D wave equation

Open questions

More open questions than closed ones!

Open question #1

Backstepping and saturating control?

Open question #2

Add dynamics to improve the performance?

[Mattioni, Le Gorrec, Zwart; 2022] and [Vanspranghe, Ferrante and CP; 2023]

Open question #3

Other **nonlinearities** (as backlash operators) and other PDEs

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Acknowledgment

Thanks for your attention

Support from the Chair on *AI and Automatic Control* and



...and support of all my colleagues and students.

Merci beaucoup

